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Low power Modes for STM32F407

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Mode Low power du STM32F407

En fonctionnement classique, le STM32F407 fonctionne en mode run: Il existe 3 mode de fonctionnement basse consommation:

- Sleep Mode
- Stop Mode
- Standby Mode

L'utilisation de ces modes de consommation est subordonnée à différents mécanismes:

- WFI: Wait for Interrupt
- WFE: Wait for Event
- Sleep on exit

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Ces modes découlent de bits de configurations du registre PWR_CR

5.4.1 PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx

Address offset: 0x00

Reset value: 0x0000 4000 (reset by wakeup from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	VOS	Reserved				FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	rw					rw	rw	rw	rw	rw	rw	w	w	rw	rw

Bit 1 **PDDS**: Power-down deepsleep

This bit is set and cleared by software. It works together with the LPDS bit.

0: Enter Stop mode when the CPU enters deepsleep. The regulator status depends on the LPDS bit.

1: Enter Standby mode when the CPU enters deepsleep.

Bit 0 **LPDS**: Low-power deepsleep

This bit is set and cleared by software. It works together with the PDDS bit.

0: Voltage regulator on during Stop mode

1: Voltage regulator in low-power mode during Stop mode

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Résumé du fonctionnement des 3 modes et de leur effets:

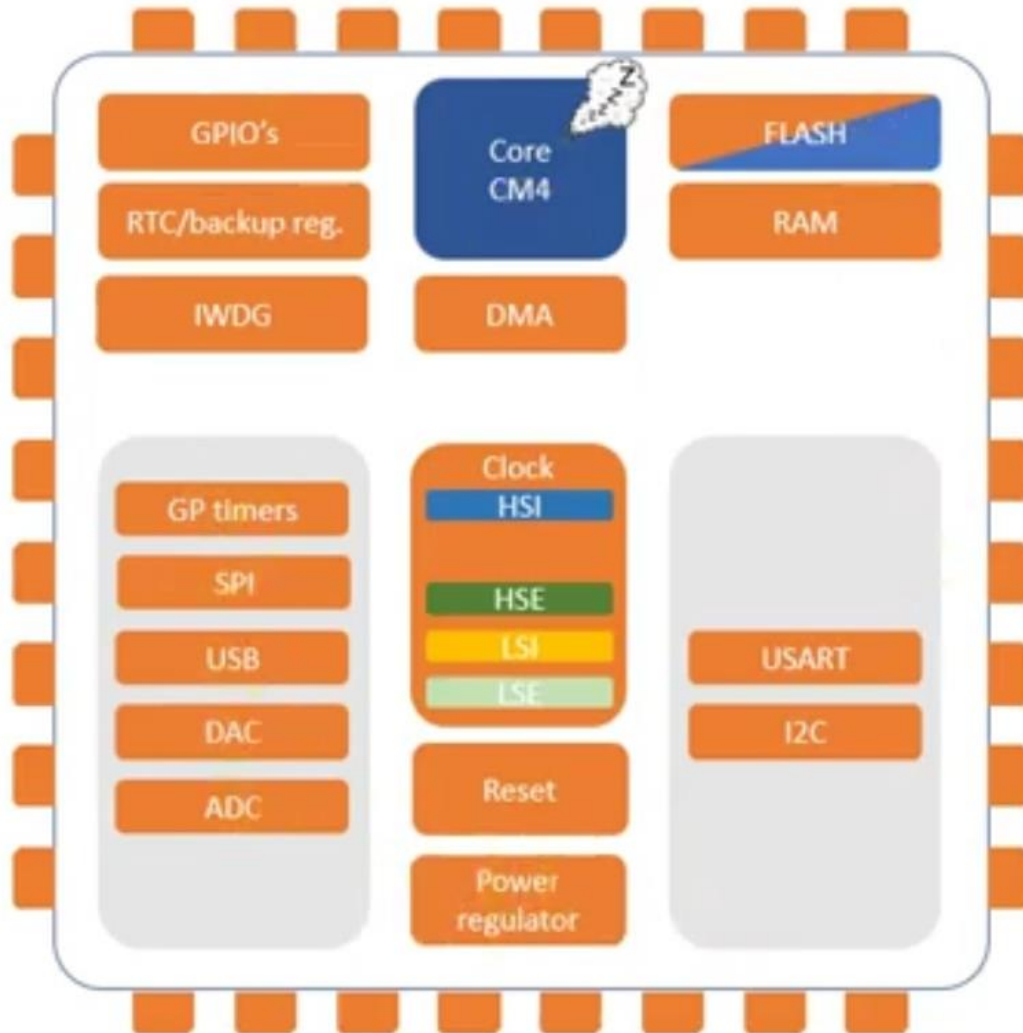
Table 23. Low-power mode summary

Mode name	Entry	Wakeup	Effect on 1.2 V domain clocks	Effect on V _{DD} domain clocks	Voltage regulator
Sleep (Sleep now or Sleep-on-exit)	WFI or Return from ISR	Any interrupt	CPU CLK OFF no effect on other clocks or analog clock sources	None	ON
	WFE	Wakeup event			
Stop	PDDS and LPDS bits + SLEEPDEEP bit + WFI, Return from ISR or WFE	Any EXTI line (configured in the EXTI registers, internal and external lines)	All 1.2 V domain clocks OFF	HSI and HSE oscillators OFF	ON or in low-power mode (depends on <i>PWR power control register (PWR_CR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> and <i>PWR power control register (PWR_CR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> <i>xPWR power control register (PWR_CR)</i> for <i>STM32F42xxx</i> and <i>STM32F43xxx</i>)
Standby	PDDS bit + SLEEPDEEP bit + WFI, Return from ISR or WFE	WKUP pin rising edge, RTC alarm (Alarm A or Alarm B), RTC Wakeup event, RTC tamper events, RTC time stamp event, external reset in NRST pin, IWDG reset			OFF

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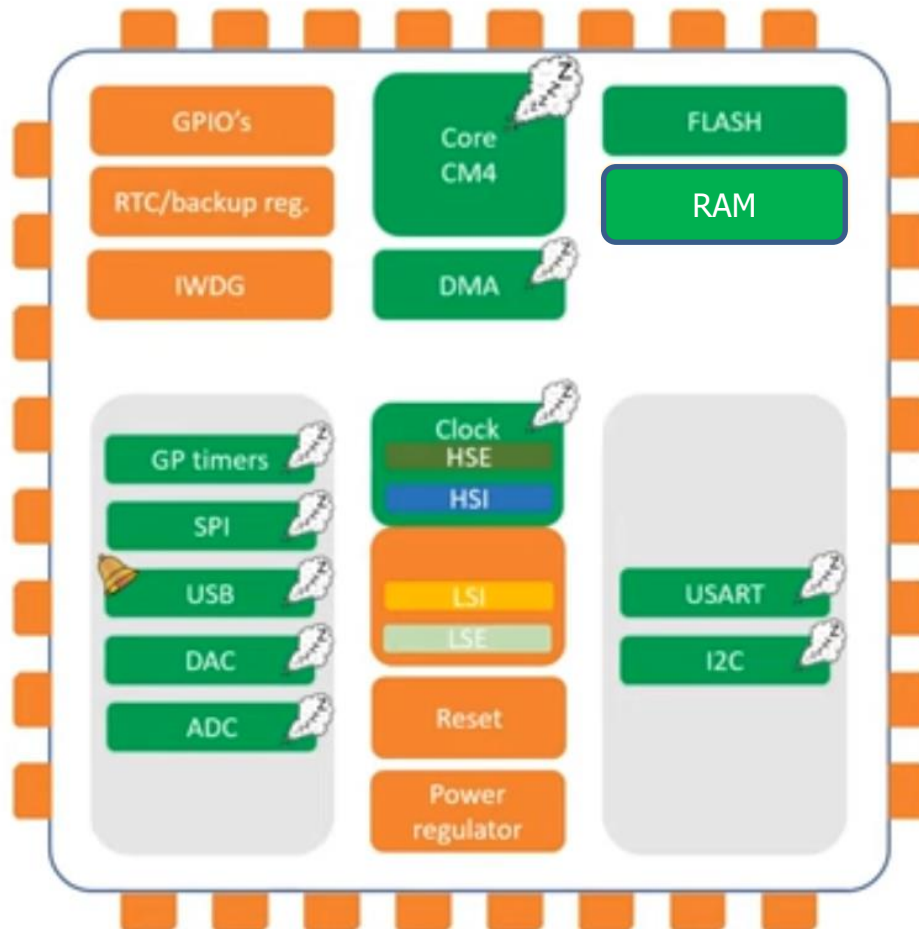
SLEEP Mode



- **Core is stopped**
- **Peripherals are running**

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STANDBY MODE



- **Core is stopped**
- **HSE, MSI clocks are OFF**
- **SRAM datas are lost (except backup domain)**
- **Peripherals are OFF (except RTC)**



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Consommation des 3 modes :

